

## Human Review: $\tau_\infty$ -Density Without $p$ -OAP

Original automatic proof: `solution_packet.pdf`

Checked date: 14 June 2026

Status: Rejected after a second AI pass; not independently verified by the human reviewer.

Verdict: This packet should not be treated as a verified solution. A second AI review identified a serious gap in the key density lemma. The human reviewer did not directly verify the full argument independently.

## Reason for Rejection

The proposed proof rests on the lemma claiming that, for arbitrary operator spaces  $W, V$ ,

$$\mathcal{F}(W, V)$$

is  $\tau_{cc}$ -dense in  $\mathcal{CB}(W, V)$ , and hence also  $\tau_\infty$ -dense.

The second AI pass identified a gap in the proof of this lemma. The argument chooses, for a completely compact matrix set, a finite-dimensional subspace  $E \subseteq W$  which approximates the test set within some tolerance  $\delta$ . It then chooses a finite-rank completely bounded map

$$P : W \rightarrow E$$

with  $P|_E = \text{id}_E$ , and estimates

$$\|(T - T \circ P)_n x\| \leq \|T\|_{cb}(1 + \|P\|_{cb})\delta.$$

The problem is that the projection  $P$ , and especially its completely bounded norm  $\|P\|_{cb}$ , depends on the finite-dimensional space  $E$ . But  $E$  itself was chosen depending on  $\delta$ . Thus the final instruction to choose

$$\delta < \frac{\varepsilon}{\|T\|_{cb}(1 + \|P\|_{cb})}$$

is circular.

Complete compactness gives approximation by finite-dimensional subspaces, but it does not provide finite-rank projections onto those subspaces with uniformly controlled completely bounded norm. This is analogous to the classical false argument that every Banach space has the approximation property: compact sets can be approximated by finite-dimensional subspaces, but without controlled projections this does not yield finite-rank approximation of the identity.

## Scope of This Review

This note records a rejection after a second pass by the AI model. It should not be read as a direct independent human verification of all details by Antonio Acuaviva. Rather, the packet is being reclassified because the second AI pass found a serious and plausible obstruction in the central lemma.

## Review Outcome

Rejected. The proposed solution does not currently prove the claimed answer to the source question. Unless the projection-norm problem can be repaired by an additional argument or by a different construction, this packet should be excluded from the verified examples.